# VECTORS AND VECTOR SPACES

## A Detailed Foundation for Quantum Computing

From geometric vectors to complex state spaces, bases, normalization, multi-qubit dimensions, and tensor products

**Quantum Mathematics Roadmap - Topic 2**

Prepared as a standalone reference volume

# Contents



**How to use this volume:** Read Sections 1-10 for the mathematical foundation. Sections 11-13 connect the ideas directly to quantum states. Use the worked examples and exercises as a revision checkpoint.

## Learning objectives

- Interpret a vector as both an ordered list and a geometric object.
- Perform vector addition, scalar multiplication, and linear combinations.
- Understand the formal structure of a vector space and subspace.
- Determine span, linear dependence, basis, coordinates, and dimension.
- Translate between mathematical vectors and quantum ket notation.
- Explain why one qubit lives in C^2 and n qubits live in C^(2^n).
- Use tensor products to construct multi-qubit basis states.

# 1. Why vectors are the language of quantum states

A complex number can represent one quantum amplitude. A full quantum state usually contains several amplitudes at once, one for each basis state. A vector is the natural object for storing this ordered collection without losing which amplitude belongs to which basis state.

$$|psi> = alpha|0> + beta|1> \quad <=> \quad |psi> = [alpha, beta]^T$$

The first component is the amplitude associated with |0>, and the second component is the amplitude associated with |1>. The order is essential: swapping the components generally changes the state.

**Central viewpoint:** Quantum states are vectors; quantum gates are linear transformations acting on those vectors; measurements extract information from their components relative to a chosen basis.

A one-qubit state is a normalized complex vector

|0> = [1, 0]^T

|1> = [0, 1]^T

|psi> = alpha|0> + beta|1>

**Column form:**

|psi> = [alpha, beta]^T

**Measurement:**

P(0)=|alpha|^2,  P(1)=|beta|^2

|alpha|^2 + |beta|^2 = 1

*Figure 1. A one-qubit state represented as a normalized complex column vector.*

## 1.1 Algebraic and geometric viewpoints

A vector has two useful interpretations. Algebraically, it is an ordered list of scalars. Geometrically, a real two-dimensional vector can be pictured as an arrow from the origin. The algebraic interpretation generalizes cleanly to high-dimensional and complex spaces, even when a direct picture is no longer possible.

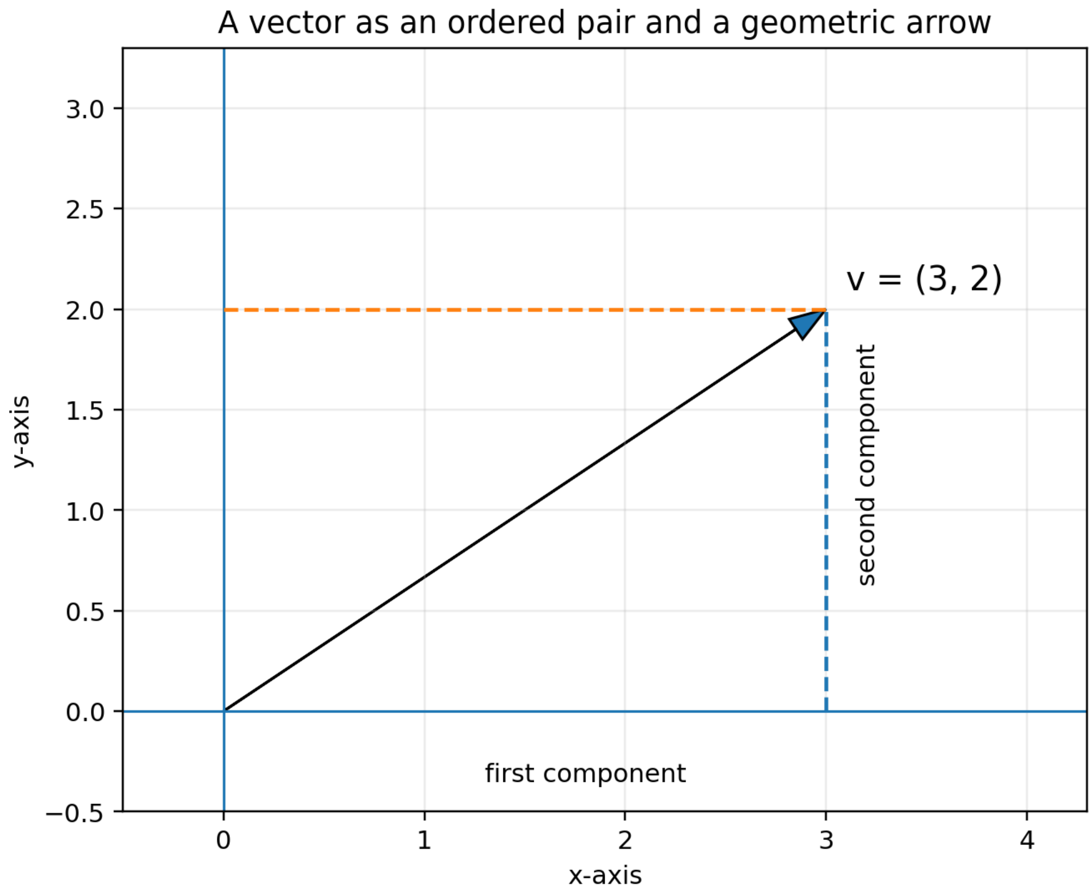


*Figure 2. The vector (3, 2)^T as an ordered pair and an arrow.*

In quantum computing, the geometric picture remains valuable for intuition, but the algebraic representation is the one used in calculations. A one-qubit vector has two complex components, so it contains four real numbers before normalization and global-phase redundancy are considered.

# 2. Scalars, vectors, components, and notation

## 2.1 Scalars

A scalar is a single element of the number system over which the vector space is defined. In an ordinary real vector space, scalars belong to R. In quantum mechanics, scalars belong to C because amplitudes may contain phase.

| Space | Allowed scalars | Typical vector |
|---|---|---|
| R^2 | Real numbers | [x, y]^T |
| C^2 | Complex numbers | [alpha, beta]^T |
| C^n | Complex numbers | [v1, ..., vn]^T |

## 2.2 Vectors and components

v = [v1, v2, ..., vn]^T

The entries $v_1, v_2, \ldots, v_n$ are called components or coordinates. The superscript T means transpose: a row has been turned into a column. Quantum state vectors are conventionally written as columns because matrices act on them from the left.

**Order matters:** $[1, i]^T$ and $[i, 1]^T$ contain the same two numbers but represent different vectors and generally different quantum states.

### 2.3 Equality of vectors

Two vectors are equal only when they have the same dimension and all corresponding components are equal.

$$[a, b]^T = [c, d]^T \quad \text{iff} \quad a = c \text{ and } b = d$$

## 3. Geometric interpretation and basic operations

### 3.1 Vector addition

Vectors of the same dimension are added component by component. Geometrically, the same rule is represented by the tip-to-tail or parallelogram construction.

$$[u_1, u_2]^T + [v_1, v_2]^T = [u_1 + v_1, u_2 + v_2]^T$$

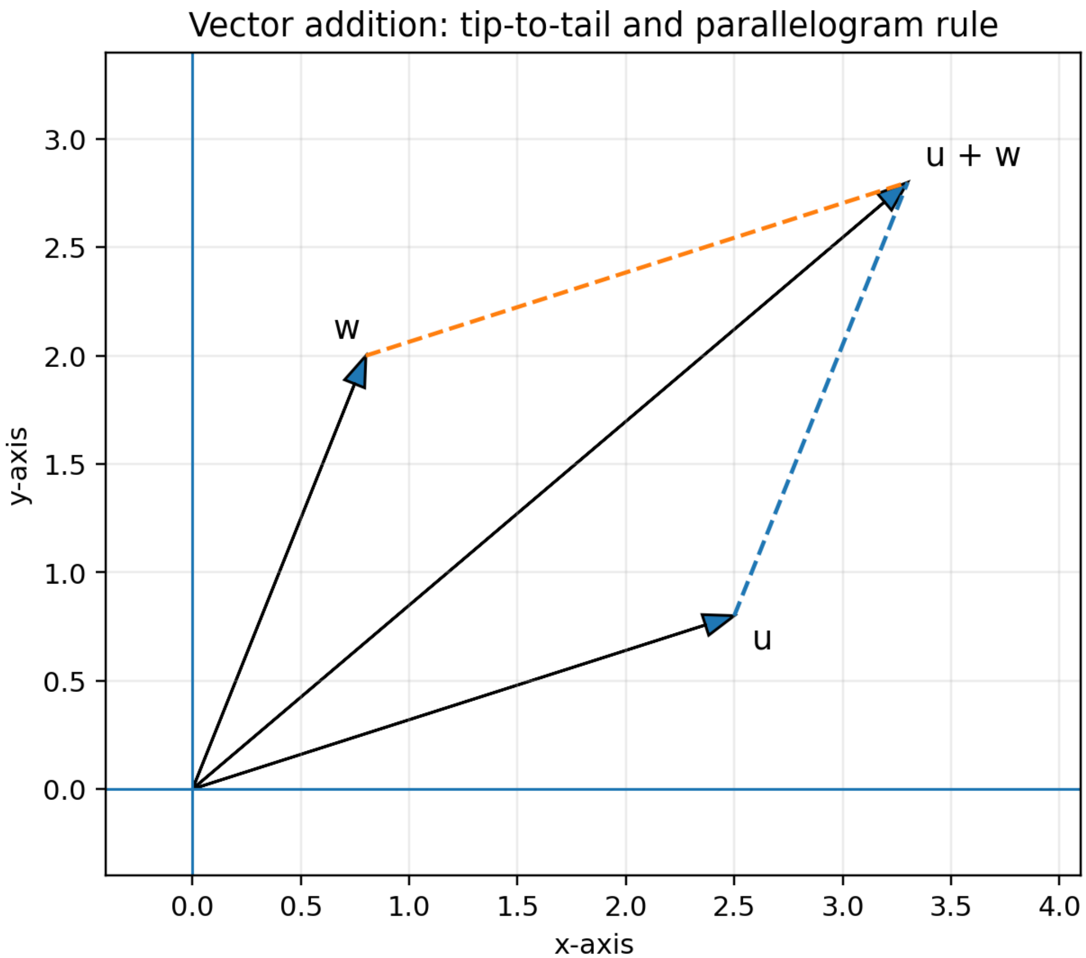


*Figure 3. Vector addition using the parallelogram rule.*

Addition is fundamental to quantum superposition. The expression alpha|0> + beta|1> is a vector sum after the basis vectors have been scaled by their amplitudes.

## 3.2 Scalar multiplication

Multiplying a vector by a scalar multiplies every component by the same scalar.

$$c[v1, v2, ..., vn]^T = [cv1, cv2, ..., cvn]^T$$

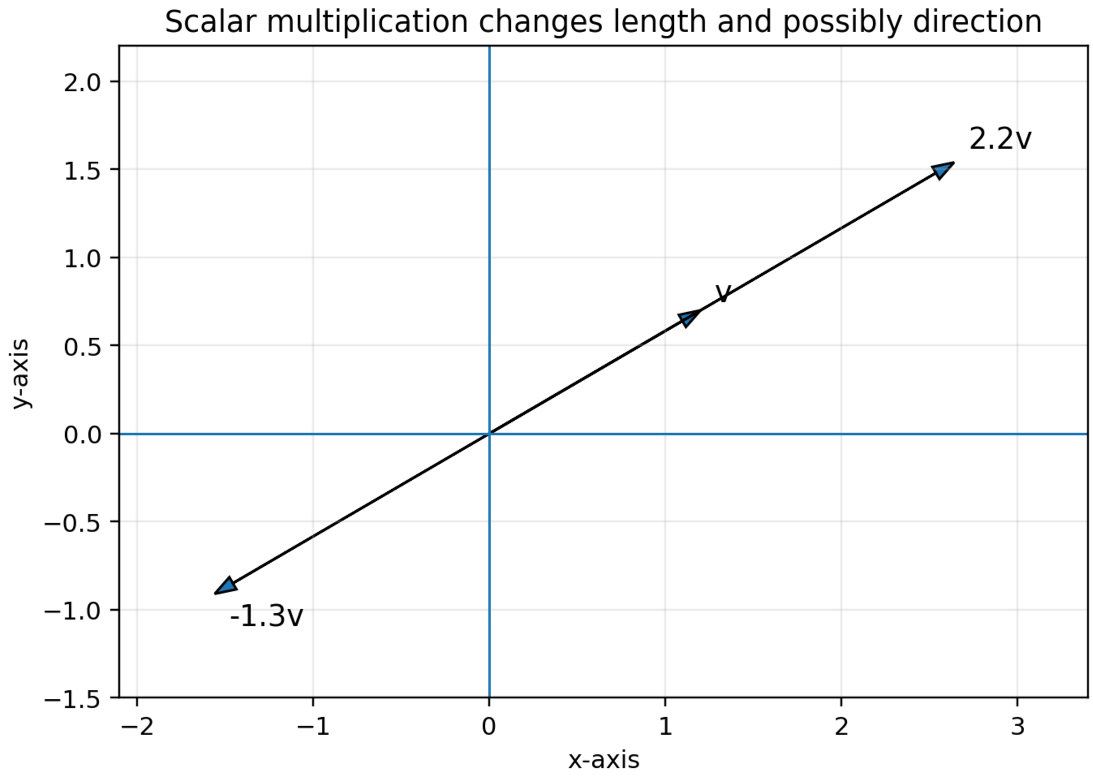


*Figure 4. Positive and negative scalar multiplication.*

For real geometric vectors, a positive scalar changes length, while a negative scalar also reverses direction. For complex vectors, the scalar may additionally change phase. Multiplying an entire normalized quantum state by e^(i gamma) produces a global phase and does not change the physical pure state.

## 3.3 Algebraic properties

| Property | Statement |
|---|---|
| Commutative addition | u + v = v + u |
| Associative addition | (u + v) + w = u + (v + w) |
| Additive identity | v + 0 = v |
| Additive inverse | v + (-v) = 0 |
| Distributivity | c(u + v) = cu + cv |
| Scalar distribution | (a + b)v = av + bv |
| Scalar associativity | a(bv) = (ab)v |
| Scalar identity | 1v = v |

# 4. Linear combinations and vector-space axioms

## 4.1 Linear combinations

A linear combination is formed by scaling vectors and adding the results.

$$a1\ v1 + a2\ v2 + \dots + ak\ vk$$

The coefficients a1, ..., ak come from the underlying scalar field. Over C, complex coefficients are allowed. The idea of linear combination is the foundation of span, basis, superposition, and state decomposition.

**Quantum interpretation:** |psi> = alpha|0> + beta|1> is a linear combination of the computational-basis vectors. The amplitudes alpha and beta are its coordinates in that basis.

## 4.2 Formal definition of a vector space

A vector space V over a field F is a set equipped with vector addition and scalar multiplication that satisfy the standard axioms listed above. The field is usually R or C. The zero vector and additive inverses must exist, and the space must be closed under both operations.

Closure means that operating on valid vectors cannot take us outside the space: if u and v are in V, then u + v is in V; if c is in F and v is in V, then cv is in V.

## 4.3 Examples and non-examples

| Set | Vector space? | Reason |
|---|---|---|
| R^2 | Yes | Closed under real addition and real scalar multiplication. |
| C^2 over C | Yes | Closed under complex scalars and addition. |
| C^2 over R | Yes | It can also be treated as a real vector space of real dimension four. |
| Unit vectors only | No | Adding two unit vectors need not produce a unit vector. |
| Normalized quantum states only | No | Not closed under addition or arbitrary scalar multiplication. |
| All 2x2 complex matrices | Yes | Matrix addition and scalar multiplication satisfy the axioms. |

**Important subtlety:** The set of normalized quantum states is not itself a vector space, even though every state is represented by a vector in a larger vector space. Normalization restricts us to a curved subset of that space.

# 5. Subspaces and solution spaces

## 5.1 Subspace definition

A subspace W of a vector space V is a subset that is itself a vector space using the same addition and scalar multiplication. A practical subspace test checks three conditions:

17. The zero vector belongs to W.
18. If u and v belong to W, then u + v belongs to W.
19. If v belongs to W and c is a scalar, then cv belongs to W.

A line through the origin in $R^2$ is a subspace. A line that does not pass through the origin is not, because it does not contain the zero vector and fails closure under scalar multiplication.

## 5.2 Null spaces as subspaces

For a matrix A, the set of all vectors satisfying Ax = 0 is called the null space or kernel of A. It is always a subspace because linear combinations of solutions remain solutions:

$$A(au + bv) = aAu + bAv = 0$$

Null spaces become important when studying linear dependence, eigenvectors, quantum error-correcting codes, stabilizers, and constraint systems.

# 6. Span and generating sets

## 6.1 Definition of span

The span of vectors v1, ..., vk is the set of all their linear combinations.

$$\mathrm{span}\{v1, \ldots, vk\} = \{a1v1 + \ldots + akvk : ai \text{ in } F\}$$

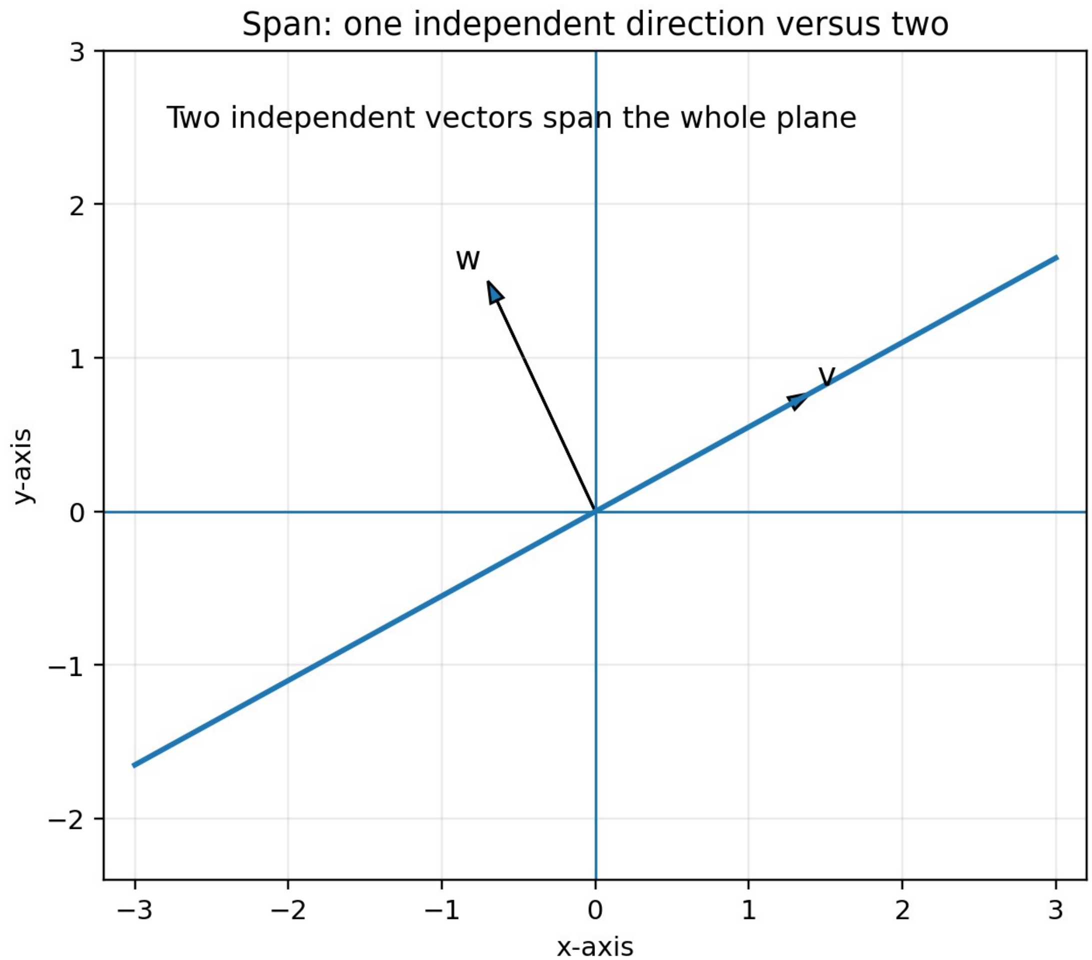


*Figure 5. One nonzero vector spans a line; two independent vectors span a plane.*

A spanning set is a collection of vectors capable of generating the entire target space. It may contain redundant vectors; a basis is a spanning set with all redundancy removed.

## 6.2 Testing whether a vector belongs to a span

To determine whether b lies in span{v1, ..., vk}, solve the linear equation

$$a1v1 + \dots + akvk = b$$

If the coefficient system has a solution, b lies in the span. In matrix form, place the generating vectors as columns of A and solve Ax = b.

## 6.3 Worked example

Determine whether b = (5, 1)^T lies in the span of v1 = (1, 1)^T and v2 = (2, -1)^T.

$$a[1,1]^T + b[2,-1]^T = [5,1]^T$$

Equating components gives a + 2b = 5 and a - b = 1. The second equation gives a = 1 + b. Substituting into the first gives 1 + 3b = 5, so b = 4/3 and a = 7/3. A solution exists, so the vector lies in the span.

# 7. Linear dependence and independence

## 7.1 Formal definition

Vectors v1, ..., vk are linearly independent if the equation

$$a1v1 + a2v2 + \dots + akvk = 0$$

has only the trivial solution a1 = a2 = ... = ak = 0. If a nontrivial solution exists, the set is linearly dependent.

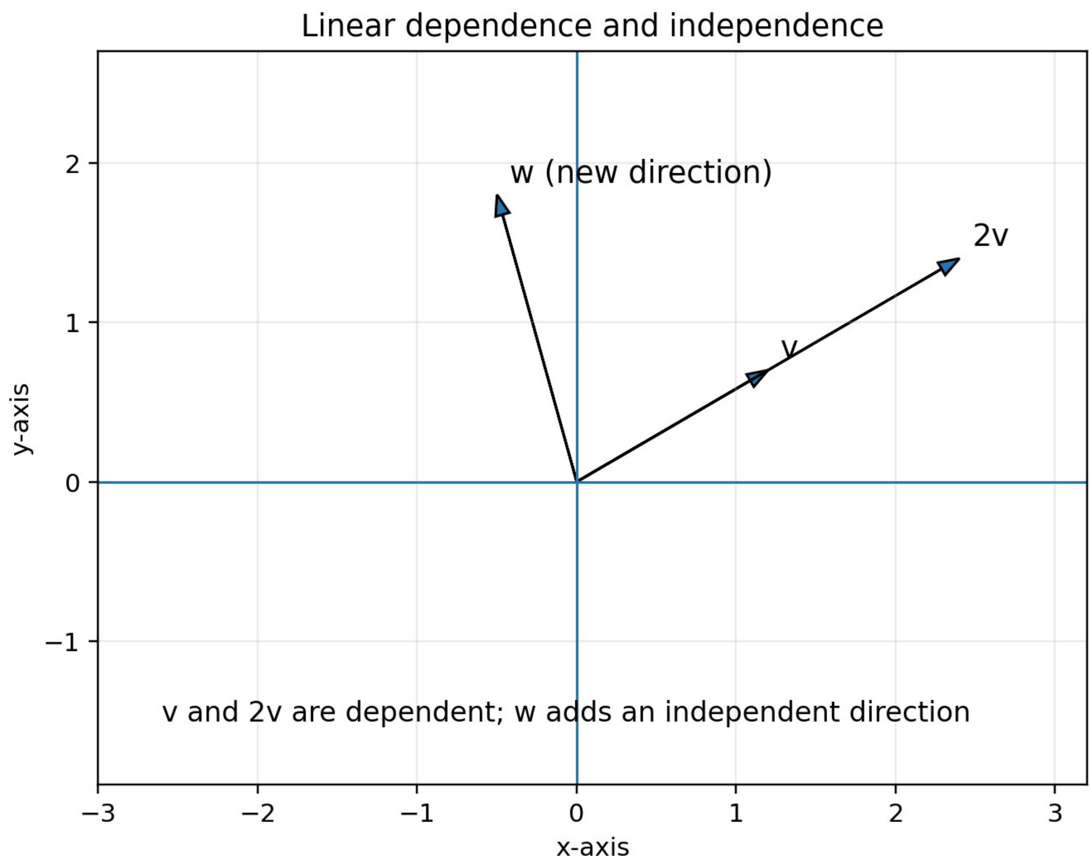


*Figure 6. Scalar multiples are dependent; an additional direction can restore independence.*

## 7.2 Practical intuition

- Two nonzero vectors in a plane are dependent exactly when one is a scalar multiple of the other.
- Any set containing the zero vector is dependent.
- More than n vectors in an n-dimensional vector space must be dependent.
- A dependent spanning set contains at least one removable vector.

## 7.3 Matrix test

Place the vectors as columns of a matrix A. The columns are independent exactly when Ax = 0 has only the zero solution. For a square matrix, this is equivalent to det(A) not equal to zero.

**Why independence matters:** Independent vectors encode genuinely separate directions or degrees of freedom. Basis vectors must be independent so that coordinates are unique.

# 8. Bases, coordinates, and dimension

## 8.1 Basis definition

A basis of V is a set of vectors that is both linearly independent and spanning. These two requirements provide exactly enough directions to describe every vector uniquely.

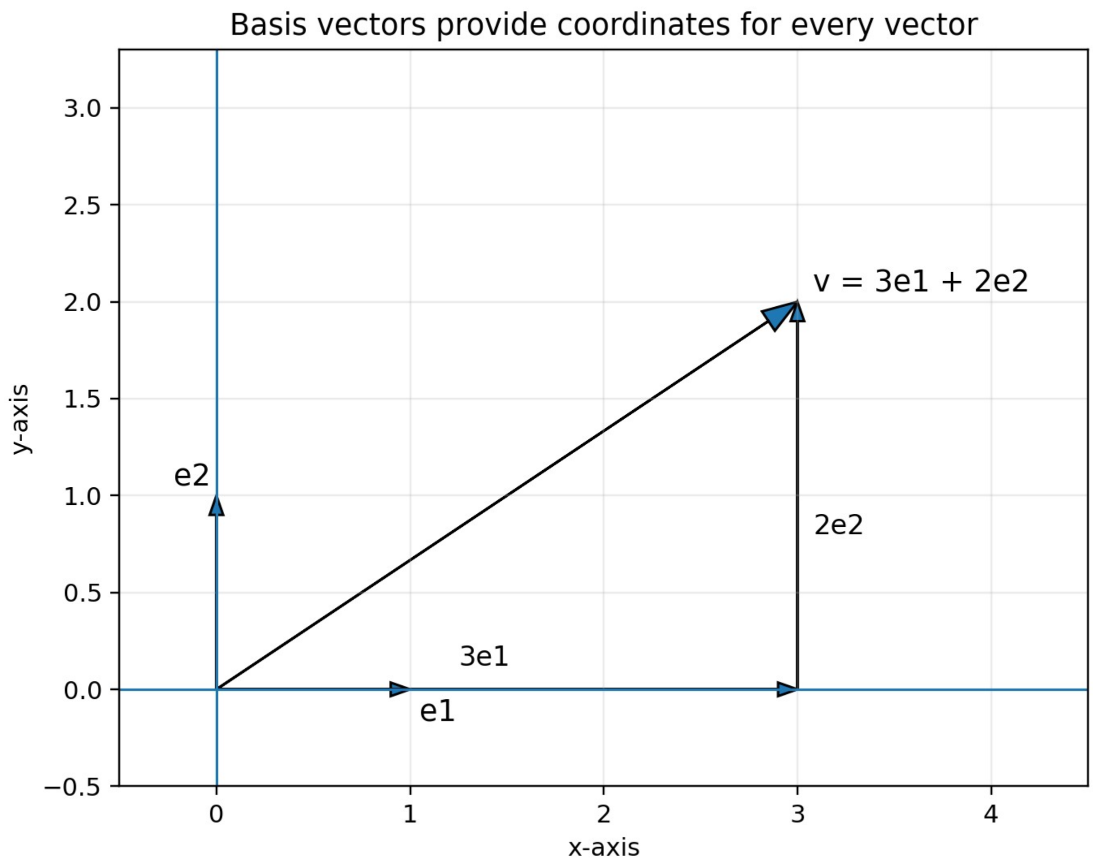


*Figure 7. Coordinates arise from decomposition in a chosen basis.*

$$v = c_1 b_1 + c_2 b_2 + \dots + c_n b_n$$

The coefficients $c_1, \dots, c_n$ are the coordinates of v relative to the basis $B = \{b_1, \dots, b_n\}$. Coordinate uniqueness follows from linear independence.

## 8.2 Standard and nonstandard bases

The standard basis of $C^2$ is $e_1 = (1,0)^T$ and $e_2 = (0,1)^T$. But many other bases are possible. In quantum computing, the computational basis {|0>, |1>} and the X-basis {|+>, |->} are two different orthonormal bases of the same space.

| **Basis** | **Vectors** | **Typical role** |
|---|---|---|
| Computational / Z | \|0>, \|1> | Standard measurement and |

| Basis | Vectors | Typical role |
|---|---|---|
| | | bit representation |
| X basis | $\vert +\rangle$, $\vert -\rangle$ | Interference and X measurements |
| Y basis | $\vert +i\rangle$, $\vert -i\rangle$ | Relative phase and Y measurements |

### 8.3 Dimension

The dimension of a finite-dimensional vector space is the number of vectors in any basis. Every basis of a given finite-dimensional space contains the same number of vectors.

$$\dim(C^n) = n$$

Dimension counts independent directions, not the number of vectors in the space. $C^2$ contains infinitely many vectors but has dimension two over C. If $C^2$ is viewed as a real vector space, its real dimension is four because each complex component contains two real numbers.

### 8.4 Coordinates depend on basis

The vector itself and its coordinate column must be distinguished. A coordinate column is meaningful only after a basis has been specified. The same abstract vector can have different coordinate columns in different bases.

## 9. Change of basis

Changing basis changes the coordinate description, not the underlying vector. This is analogous to describing the same physical displacement using different coordinate axes.

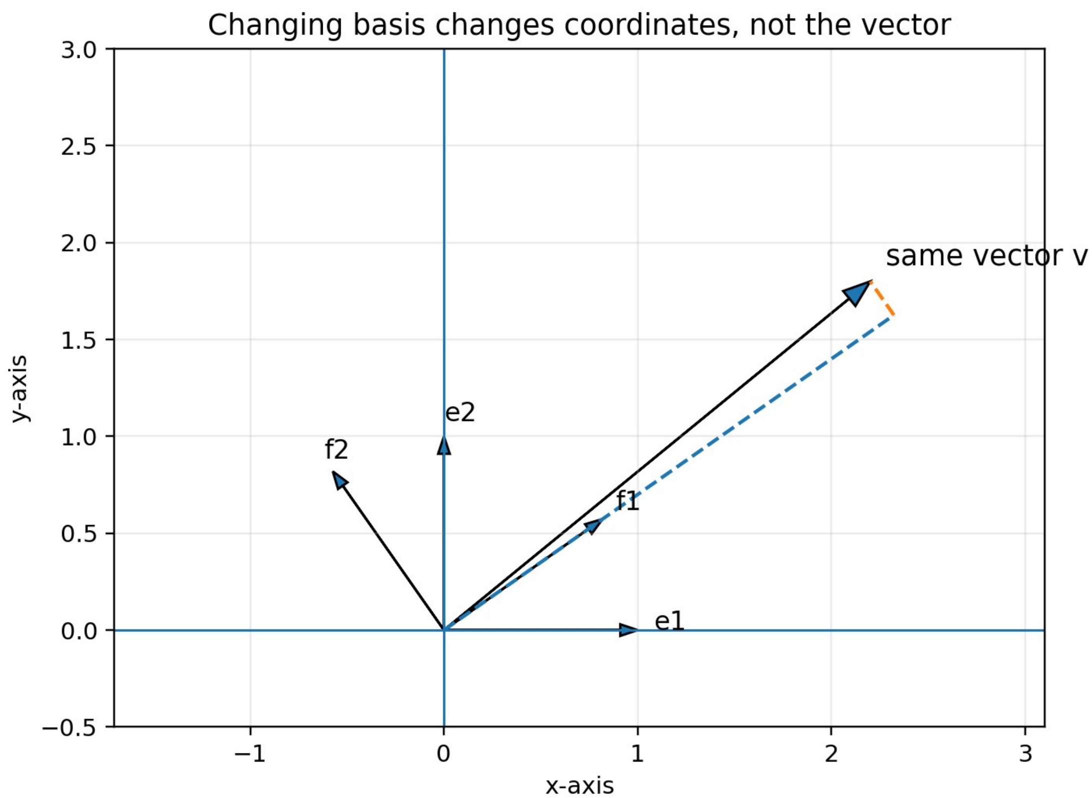


*Figure 8. The same vector has different coordinates in two bases.*

## 9.1 Change-of-basis matrix

Let B = {b1, ..., bn} be a basis, and place its basis vectors as columns of a matrix S. If [v]_B denotes the coordinate column of v in basis B, then

$$v\_standard = S\,[v]\_B$$

If S is invertible, the reverse conversion is

$$[v]\_B = S^{-1}\, v\_standard$$

## 9.2 Quantum example: computational basis and X basis

$$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}, \quad |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$$

The Hadamard matrix has |+> and |-> as its columns. It converts coordinates between the computational basis and X basis because H = H^(-1). Thus H acts both as a quantum gate and as a basis-change operation.

$$H = (1/\sqrt{2})\,[[1, 1], [1, -1]]$$

**Measurement connection:** Applying H and then measuring in the computational basis is equivalent to measuring the original state in the X basis.

# 10. Complex vector spaces

## 10.1 Why complex scalars are necessary

Real amplitudes can describe some quantum states, but complex amplitudes are necessary for general phase evolution, interference, unitary dynamics, and the structure of quantum mechanics. The relevant scalar field is therefore C.

$$C^n = \{ [z1, ..., zn]^T : zk \text{ in } C \}$$

All familiar vector operations continue to work component by component. The main change appears in notions involving length, angle, and overlap: complex conjugation is required.

## 10.2 Complex scalar multiplication and global phase

Multiplication by a complex scalar can change both magnitude and phase. For normalized quantum states, multiplication by a unit complex number e^(i gamma) changes every component by the same phase and leaves all physical probabilities unchanged.

$$|psi\rangle \text{ and } e^{(i\ gamma)}|psi\rangle \text{ represent the same physical pure state}$$

## 10.3 C^2 is not an ordinary geometric plane

Although C^2 has two complex components, it has four real degrees of freedom before constraints. It should not be confused with the ordinary two-dimensional real plane. The Bloch sphere becomes possible only after normalization and removal of global phase reduce the number of physically relevant real parameters to two.

# 11. Norms, normalization, and orthogonality preview

## 11.1 Norm

The norm measures vector length. For a complex vector, the squared magnitudes of the components are added.

$$||v|| = sqrt(|v1|^2 + |v2|^2 + ... + |vn|^2)$$

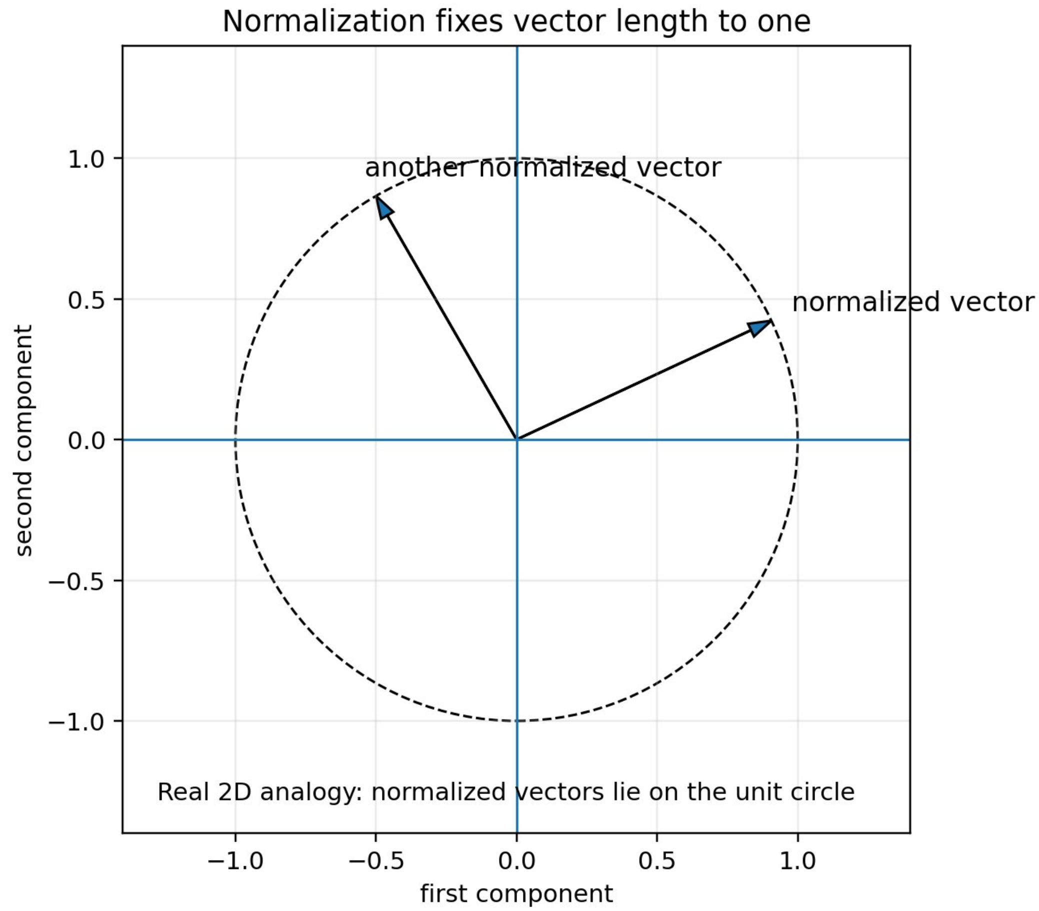


*Figure 9. Real two-dimensional analogy for normalized vectors.*

## 11.2 Normalization

A nonzero vector v is normalized by dividing by its norm:

$$v\_hat = v / ||v||$$

A valid pure quantum state must have unit norm because the squared magnitudes of all measurement amplitudes must sum to one.

$$sum\_k |alpha\_k|^2 = 1$$

## 11.3 Orthogonality

In a complex vector space, orthogonality is defined using the inner product, which conjugates the first vector. Two vectors are orthogonal when their inner product is zero. Orthonormal basis vectors are both normalized and mutually orthogonal.

$$<u|v> = 0 \quad \text{means u and v are orthogonal}$$

This volume treats orthogonality only as a preview. The complete formal treatment belongs to the inner-product topic.

# 12. Single-qubit state space

## 12.1 Computational basis

$$|0\rangle = [1,0]^T, \quad |1\rangle = [0,1]^T$$

These vectors form an orthonormal basis for $C^2$. Any one-qubit pure state can be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

## 12.2 Meaning of components

The coordinates alpha and beta are probability amplitudes, not probabilities. Measurement in the computational basis produces outcome 0 with probability $|\alpha|^2$ and outcome 1 with probability $|\beta|^2$.

The relative phase between alpha and beta affects interference and measurements in other bases, even when computational-basis probabilities remain unchanged.

## 12.3 Alternative basis coordinates

$$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}, \quad |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$$

Every one-qubit state can also be expanded as $c_{plus}|+\rangle + c_{minus}|-\rangle$. The coefficients are different because coordinates depend on the basis. A basis change does not physically rotate laboratory space; it changes which pair of orthogonal quantum alternatives is used to describe or measure the state.

# 13. Multi-qubit state spaces and tensor products

## 13.1 Exponential dimension

A single qubit has two basis states. Two qubits have four basis states, three qubits have eight, and $n$ qubits have $2^n$. Therefore, the state vector of $n$ qubits contains $2^n$ complex amplitudes.

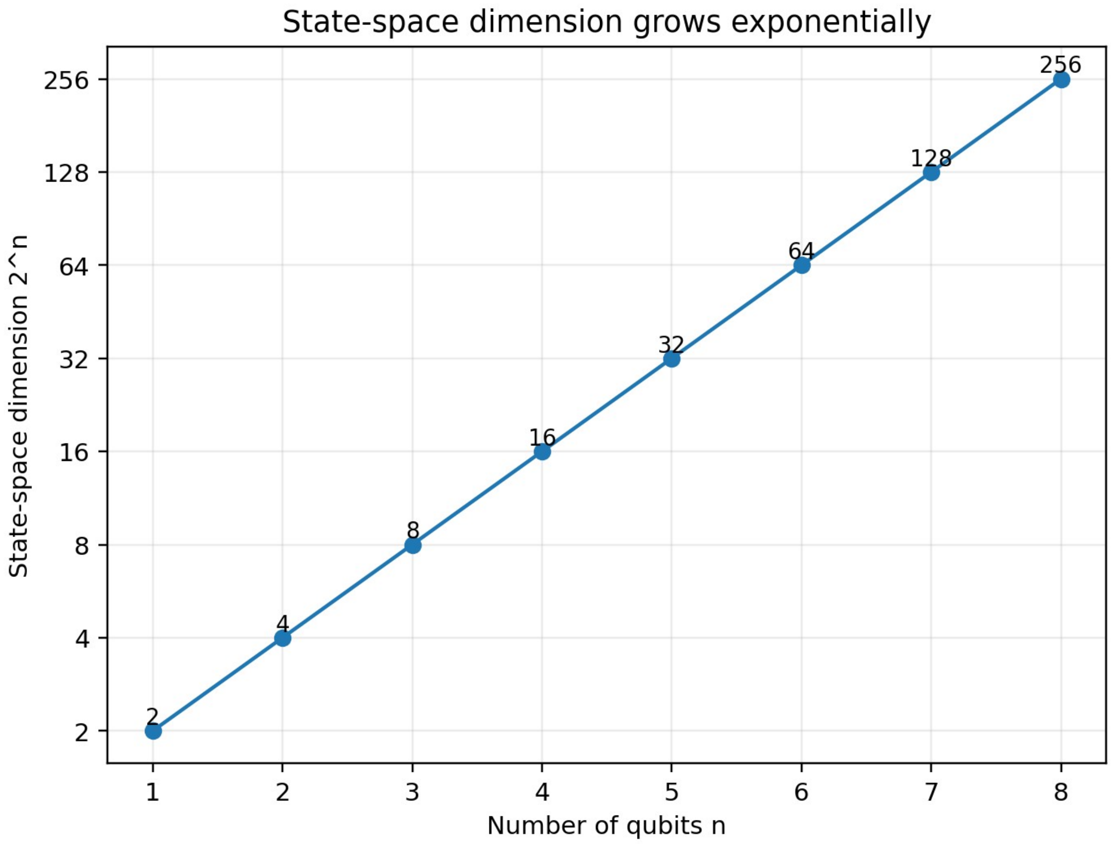


*Figure 10. Exponential growth of state-space dimension.*

dim(H_n) = 2^n

This does not mean a quantum computer stores an ordinary accessible list of 2^n classical numbers. Measurement reveals only limited information, and useful algorithms require carefully structured transformations and interference. Nevertheless, the mathematical state space grows exponentially.

## 13.2 Tensor product

Independent quantum systems are combined using the tensor product, written with the symbol x or ⊗. If |a> has m components and |b> has n components, then |a> ⊗ |b> has mn components.

[a1, a2]^T tensor [b1, b2]^T = [a1b1, a1b2, a2b1, a2b2]^T

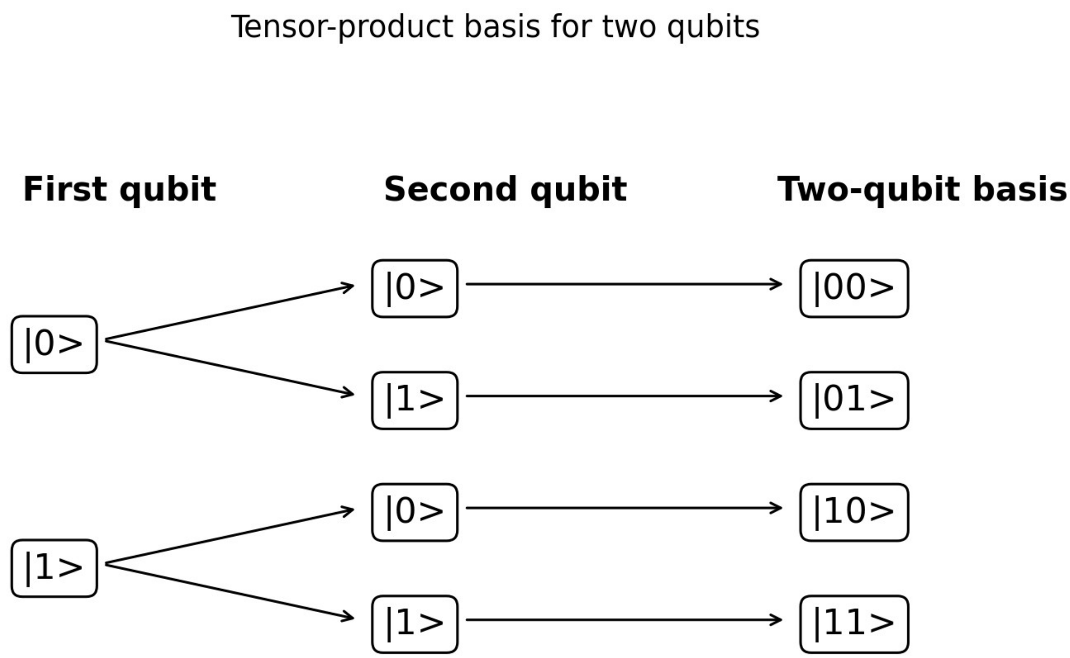


*Figure 11. Tensor products generate the four computational-basis states of two qubits.*

## 13.3 Two-qubit computational basis

$$|00>, |01>, |10>, |11>$$

| State | Tensor form | Column vector |
|---|---|---|
| \|00> | \|0> tensor \|0> | [1,0,0,0]^T |
| \|01> | \|0> tensor \|1> | [0,1,0,0]^T |
| \|10> | \|1> tensor \|0> | [0,0,1,0]^T |
| \|11> | \|1> tensor \|1> | [0,0,0,1]^T |

## 13.4 General two-qubit state

$$|psi> = a|00> + b|01> + c|10> + d|11>$$

$$|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$$

Some two-qubit states can be written as a tensor product of two one-qubit vectors. Others cannot. States that cannot be factored are entangled. The vector-space and tensor-product structure is therefore the mathematical foundation of entanglement.

### 13.5 Product-state example

Let |+> = (|0> + |1>)/sqrt(2). Then

$$|+\rangle \text{ tensor } |0\rangle = (|00\rangle + |10\rangle)/\text{sqrt}(2)$$

The first qubit is in an equal superposition, while the second qubit is definitely 0. The ordering convention used here is |q1 q2>; software frameworks may display bitstrings in an order that requires careful interpretation.

# 14. Worked examples and common mistakes

## 14.1 Vector operations with complex components

Let u = (1+i, 2)^T and v = (-i, 1-i)^T. Then

$$u + v = [1, 3-i]^T$$

For c = 2-i,

$$cu = [(2-i)(1+i), 2(2-i)]^T = [3+i, 4-2i]^T$$

## 14.2 Dependence test

Consider v1 = (1,2,3)^T and v2 = (2,4,6)^T. Since v2 = 2v1, the vectors are dependent. They span only one line through the origin, even though they live in R^3.

## 14.3 Basis verification

Consider b1 = (1,1)^T and b2 = (1,-1)^T. To test independence, solve a b1 + b b2 = 0. The component equations are a+b=0 and a-b=0, implying a=b=0. Since two independent vectors in a two-dimensional space automatically span it, they form a basis.

## 14.4 Coordinates in a nonstandard basis

Express v = (5,1)^T in the basis B = {(1,1)^T, (1,-1)^T}. Solve

$$c1[1,1]^T + c2[1,-1]^T = [5,1]^T$$

The equations are c1+c2=5 and c1-c2=1. Adding gives 2c1=6, so c1=3. Then c2=2. Thus [v]_B = (3,2)^T.

## 14.5 Normalizing a complex vector

Let v = (1+i, 2)^T. Its squared norm is |1+i|^2 + |2|^2 = 2 + 4 = 6. Therefore

$$\hat{v} = (1/\sqrt{6})[1+i, 2]^T$$

## 14.6 Tensor-product calculation

Let |a> = (alpha, beta)^T and |b> = (gamma, delta)^T. Then

$$|a\rangle \otimes |b\rangle = [\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta]^T$$

The order of these components corresponds to |00>, |01>, |10>, |11> under the chosen basis ordering.

## 14.7 Common mistakes

| Mistake | Correction |
|---|---|
| Treating vectors as unordered sets | Components are ordered and tied to basis |

| Mistake | Correction |
| --- | --- |
| | positions. |
| Calling normalized states a vector space | Normalization is not closed under addition or arbitrary scaling. |
| Confusing span with independence | A set can span while containing redundant vectors. |
| Assuming coordinates are absolute | Coordinates depend on the chosen basis. |
| Thinking C^2 is an ordinary 2D real plane | C^2 has four real dimensions before quantum-state constraints. |
| Using direct sums instead of tensor products for combined qubits | Composite quantum systems use tensor products. |
| Assuming 3 qubits need 3 amplitudes | They require 2^3 = 8 amplitudes. |

# 15. Exercises

## A. Basic vector operations

20. Compute (2, -1)^T + (3, 4)^T.
21. Compute (2-i)(1, i)^T.
22. Find x and y if (x+1, 2y)^T = (4, 6)^T.
23. Write (3, -2, 5)^T as a linear combination of the standard basis of R^3.
24. Explain why vectors of different dimensions cannot be added.

## B. Vector spaces and subspaces

25. Is the set {(x,y)^T : y = 2x} a subspace of R^2? Justify using the subspace test.
26. Is the set {(x,y)^T : y = 2x + 1} a subspace of R^2? Explain.
27. Show that the set of solutions of Ax = 0 is closed under linear combinations.
28. Explain why the set of all normalized vectors in C^2 is not a vector space.

## C. Span and independence

29. Determine whether (4,1)^T lies in span{(1,1)^T, (2,-1)^T}.
30. Determine whether {(1,2)^T, (3,6)^T} is independent.
31. Determine whether {(1,0,1)^T, (0,1,1)^T, (1,1,2)^T} is independent.
32. Can three vectors in C^2 be linearly independent? Explain without calculation.

## D. Bases and coordinates

33. Show that {(1,1)^T, (1,-1)^T} is a basis of R^2.
34. Find the coordinates of (7,1)^T in the basis {(1,1)^T, (1,-1)^T}.
35. Find the standard-coordinate matrix whose columns are the basis vectors (1,i)^T and (1,-i)^T.
36. Explain why every basis of C^4 contains exactly four vectors.

## E. Norms and quantum states

37. Normalize (3,4)^T.
38. Normalize (1+i, 1-i)^T.
39. Determine whether (sqrt(3)/2, i/2)^T is normalized.
40. Write (|0> - i|1>)/sqrt(2) as a column vector.
41. Write (sqrt(3)/2, -1/2)^T in ket notation and give computational-basis probabilities.

## F. Multi-qubit vectors and tensor products

42. How many amplitudes are required for 5 qubits?
43. Compute |0> tensor |1>.
44. Compute |+> tensor |1> and express the result in the computational basis.
45. Compute (alpha|0>+beta|1>) tensor (gamma|0>+delta|1>).
46. Determine whether (|00>+|10>)/sqrt(2) is a product state and factor it if possible.
47. Explain why the Bell state (|00>+|11>)/sqrt(2) cannot be written as a product of two one-qubit states.

# 16. Complete solutions and reference sheet

## 16.1 Selected exercise solutions

**A1.** (2,-1)^T + (3,4)^T = (5,3)^T.

**A2.** (2-i)(1,i)^T = (2-i, 1+2i)^T because (2-i)i = 2i-i^2 = 1+2i.

**B1.** Yes. It contains zero, and sums and scalar multiples preserve y=2x.

**B2.** No. It does not contain the zero vector and scalar multiplication does not preserve the offset +1.

**C1.** Solve a+2b=4 and a-b=1. This gives b=1 and a=2, so the vector lies in the span.

**C2.** Dependent because (3,6)^T = 3(1,2)^T.

**C3.** Dependent because the third vector is the sum of the first two.

**C4.** No. C^2 has dimension two, so any set of more than two vectors is dependent.

**D2.** Solve c1+c2=7 and c1-c2=1. Thus c1=4 and c2=3.

**E1.** ||(3,4)^T||=5, so the normalized vector is (3/5,4/5)^T.

**E2.** Squared norm = 2+2=4, so the normalized vector is (1/2)(1+i,1-i)^T.

**E3.** Yes: 3/4 + 1/4 = 1.

**F1.** 2^5 = 32 amplitudes.

**F2.** |0> tensor |1> = |01> = (0,1,0,0)^T.

**F3.** |+> tensor |1> = (|01>+|11>)/sqrt(2).

**F4.** The result is alpha gamma|00> + alpha delta|01> + beta gamma|10> + beta delta|11>.

**F5.** Yes: (|00>+|10>)/sqrt(2) = |+> tensor |0>.

## 16.2 Why the Bell state is not a product state

Assume (|00>+|11>)/sqrt(2) = (a|0>+b|1>) tensor (c|0>+d|1>). Expanding the right side gives ac|00> + ad|01> + bc|10> + bd|11>. Matching coefficients requires ac and bd to be nonzero, while ad=0 and bc=0. The first pair requires a,c,b,d all nonzero, contradicting ad=0 and bc=0. Therefore no such factorization exists; the state is entangled.

## 16.3 Compact reference sheet

| Concept | Definition / formula |
|---|---|
| Vector | Ordered list [v1,...,vn]^T |
| Linear combination | a1v1 + ... + akvk |
| Span | Set of all linear combinations |
| Independent set | Only trivial coefficients produce zero |
| Basis | Independent spanning set |
| Dimension | Number of vectors in a basis |
| Coordinates | Coefficients of a vector in a chosen basis |

| **Concept** | **Definition / formula** |
|---|---|
| | |
| Norm | sqrt(sum \|vk\|^2) |
| Normalized vector | Norm equals one |
| One-qubit space | C^2 |
| n-qubit space | C^(2^n) |
| Tensor-product dimension | dim(V tensor W) = dim(V)dim(W) |

## 16.4 Final conceptual map

- Complex numbers supply amplitudes and phase.
- Vectors assemble amplitudes into complete state descriptions.
- A basis identifies which alternatives the components refer to.
- Span tells what can be generated; independence removes redundancy.
- Dimension counts independent basis directions.
- Normalization converts vector components into a valid probability distribution after squared magnitudes are taken.
- Tensor products combine quantum systems and produce exponential state-space dimension.

**Bridge to the next topic:** Matrices are linear transformations of vector spaces. In quantum computing, they describe gates, basis changes, observables, and multi-qubit operations.